Experiment - 7.1.2. Possible Combinations from Three Digits

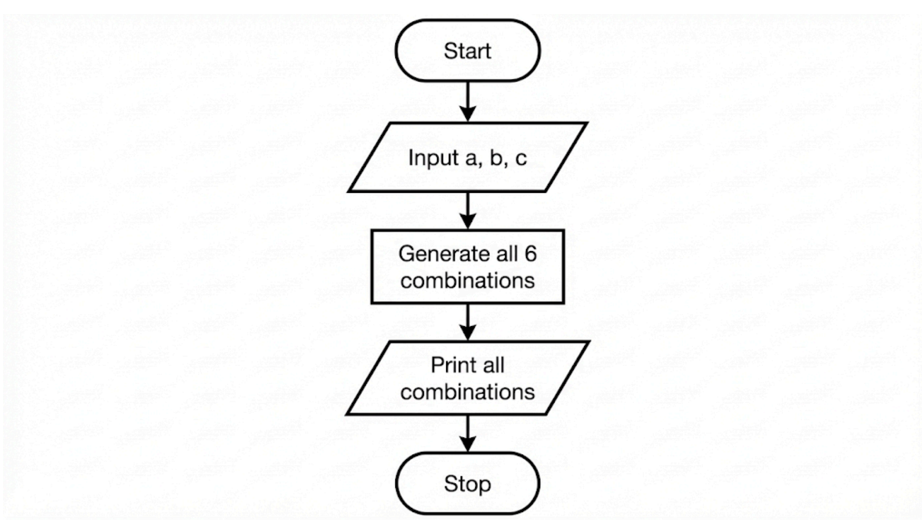
1. Aim

To design and implement a Python program that accepts three single-digit inputs (separated by spaces) and prints all six possible 3-digit combinations of these digits. The program must treat the inputs as strings to maintain any leading zeros through concatenation.

2. Pseudocode

1. **START**
2. **READ** a single line of input from the user, split it by spaces, and **STORE** the three string values into variables a, b, and c.
3. **PRINT** the concatenation of a, b, and c (representing the combination abc).
4. **PRINT** the concatenation of a, c, and b (representing the combination acb).
5. **PRINT** the concatenation of b, a, and c (representing the combination bac).
6. **PRINT** the concatenation of b, c, and a (representing the combination bca).
7. **PRINT** the concatenation of c, a, and b (representing the combination cab).
8. **PRINT** the concatenation of c, b, and a (representing the combination cba).
9. **END**

3. Flowchart



4. Python Program

```
a, b, c = input().split()

print(a + b + c)
print(a + c + b)
print(b + a + c)
print(b + c + a)
print(c + a + b)
print(c + b + a)
```

5. Experiment Screenshot